605.486 – Android Development

Module 2c
User Interaction

Accessing Views in Code

- findViewById

```
EditText nameField =
    (EditText) findViewById(R.id.name);
```

- Getting data

```
String name =
    nameField.getText().toString();
```

- Setting data

```
nameField.setText("Scott");
```

Be Careful with setText()!!!

- setText(int) expects an ID!
  - Looks up string resource!

```
int age = 42;
ageField.setText(age);
```

- *This tries to look up a string with id 42!*
- *Must explicitly convert to String!*

Button Actions (Bad Approach - 1)

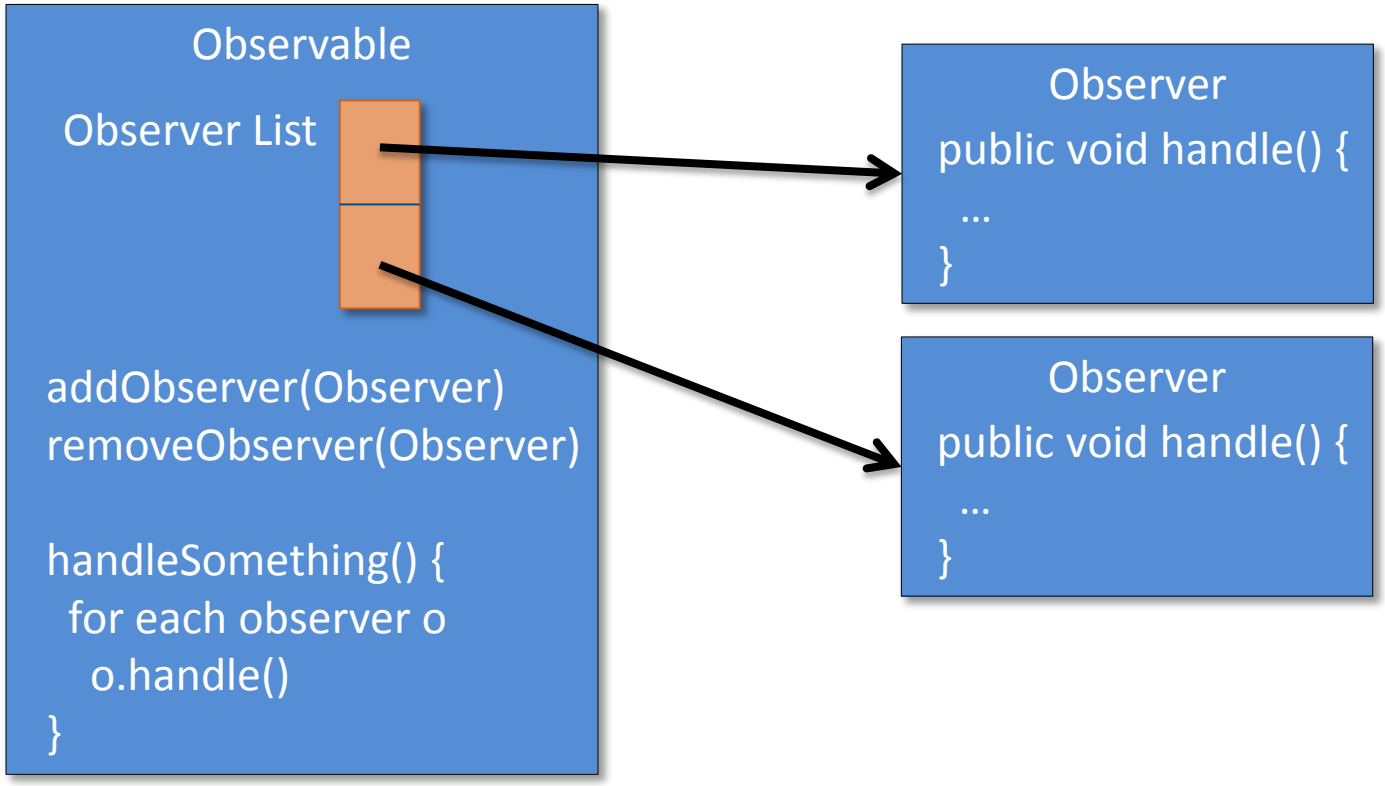
- Naive "OO" approach (not in Android)
  - Button class includes logic to do an action

```
public class SampleButton {
    …
    public void handleButtonPress() {
        // code to do something when button pressed
    }
}
```

Button Actions (Bad Approach - 2)

- Requires new button subtype for every instance
- Cannot change actions at runtime
- Cannot mix/match actions

Observer Pattern



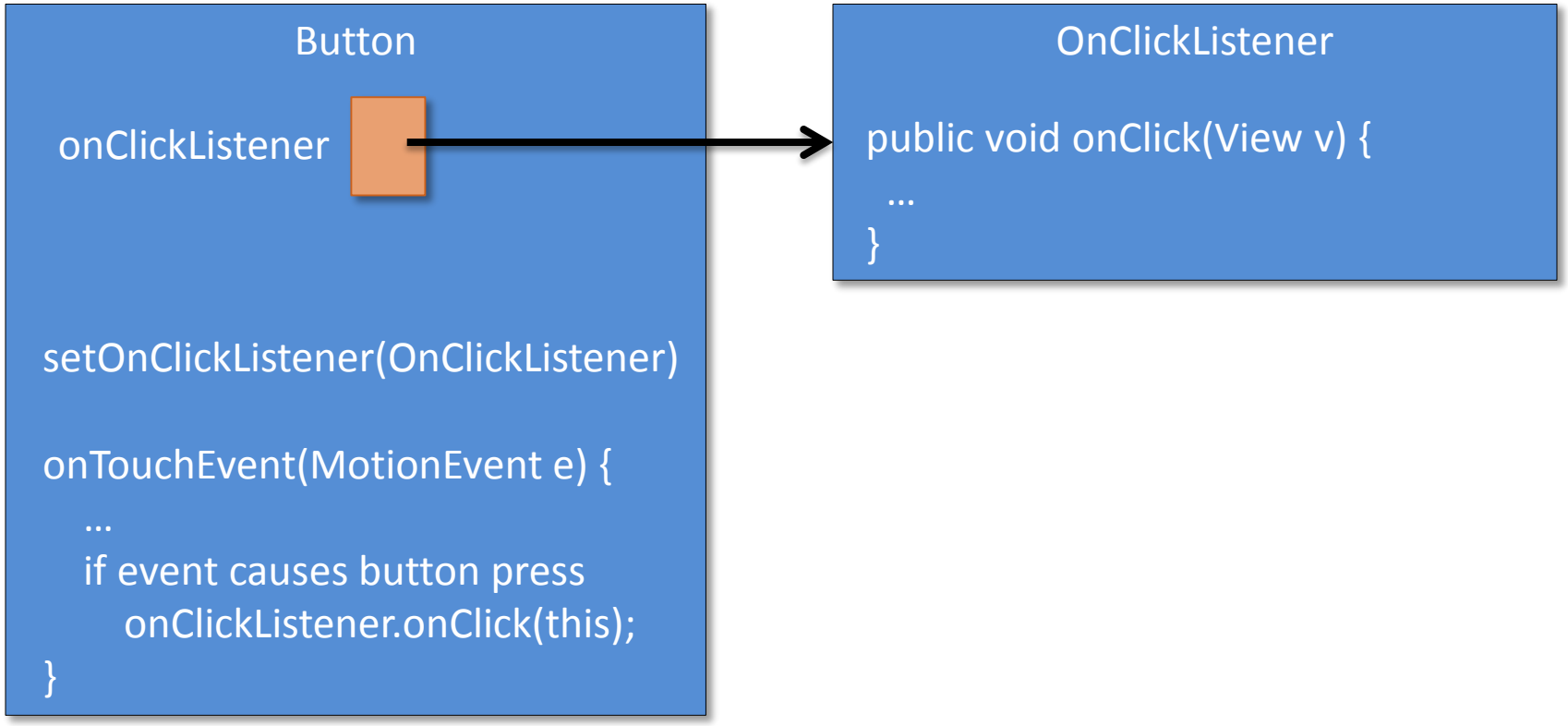
- Note: Some Observables only allow a single Observer
  - Saves memory; often only one is needed
  - setObserver() instead of addObserver()

Button Observers

- Android (and Java in general) most often use the term “listener” rather than “observer”

- Button has
  - setOnClickListener(OnClickListener)

Button Using Observer Pattern



Logging

- Log class

```
Log.v("tag", "Message");
Log.d("tag", "Message");
Log.i("tag", "Message");
Log.w("tag", "Message");
Log.e("tag", "Message");
Log.wtf("tag", "Message");

Log.e("tag", "Message", throwable);
```

- Logcat